KERALA DIPLOMA CURRICULUM REVISION 2026

Course 2182 | Semester II | Theory | CIE/ESE 40/60

Building Materials & Technology

Properties, selection, concrete, building components, masonry, foundations, roofs and construction technology.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official syllabus PDF for Course 2182 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, examples and diagrams are educational elaboration and are not presented as additional official syllabus text.

Source note: The PDF shows a teaching-scheme mismatch: 2:1:0:0 in the course-information table and 3:0:0:0 in the printed teaching/assessment row. Both official values are retained here.

COURSE DASHBOARD

At a glance

Official course information

Item	Official information
Programme	Architecture; Interior Design.
Teaching scheme	2:1:0:0 in the course-information table; 3:0:0:0 in the printed teaching/assessment row.
Contact hours	45 contact hours
Credits	3
Self-learning	4/week; 60 hours listed

Item	Official information
Prerequisite	Not separately stated in the supplied extraction.
Assessment	CIE 40 + ESE 60

Study sequence

Read the module introduction, learn each topic card, reproduce the diagram or procedure, then attempt the module questions.

Answer strategy

For a 1-mark answer define the term; for 3 marks add points; for 7 marks use a structured explanation, table, diagram or procedure.

Technical discipline

Retain official spellings, units, symbols, names, dates and distinctions when writing examination answers.

HOW TO USE THIS LESSON

A four-pass study method

Pass 1

Read the overview and identify the vocabulary of the module.

Pass 2

Study every open topic card and make a one-page notebook summary.

Pass 3

Practise the worked examples, procedures, sketches and comparisons.

Pass 4

Use the Q&A, model paper and quick revision section under timed conditions.

OUTCOMES AND ALIGNMENT

Objectives, COs and CO–PO mapping

Official course objectives

- Explain classification, properties, defects and sustainable aspects of common building materials.
- Explain cement, concrete ingredients, mix proportions, construction processes and building finishes.
- Identify building components and distinguish load-bearing/non-load-bearing elements including foundations and structural systems.
- Apply masonry, brick bonds, formwork, scaffolding, roofing and vertical-circulation knowledge in basic situations.

Course Outcomes

Course outcomes

CO	Outcome	Bloom level
CO1	Classify and explain properties, defects and sustainability of stone, timber, bamboo and eco-friendly masonry blocks.	Understand
CO2	Explain cement, concrete ingredients, mix proportions, grades, concreting, curing, defects, flooring and finishing materials.	Understand
CO3	Identify building components and distinguish load-bearing/non-load-bearing elements including foundations and structural systems.	Understand
CO4	Apply masonry, brick bonds, formwork, scaffolding, roofing and vertical-circulation knowledge in basic situations.	Apply

CO–PO matrix

CO	mapped POs
CO1	PO1=3, PO2=2, PO6=1, PO9=1, PO11=1
CO2	PO1=2, PO2=3, PO3=2, PO5=2, PO6=3, PO7=1, PO9=1, PO11=1
CO3	PO1=3, PO2=3, PO3=2, PO4=2, PO5=2, PO6=1, PO9=1, PO11=1

CO	mapped POs
CO4	PO1=2, PO2=2, PO3=3, PO4=2, PO5=1, PO6=1, PO7=1, PO8=2, PO9=2, PO10=1, PO11=1

COVERAGE CONTROL

Complete official syllabus coverage

Every official module and topic appears in the teaching cards below. Use this table as a checklist before the examination.

Module coverage

Module	Official syllabus items represented	Hours
I	Classification; stone; timber; bamboo; clay, fly ash, AAC and CSEB blocks.	7 L + 4 T
II	Cement; ingredients; PCC/RCC; M15/M20/M25; mix/w-c; concreting; curing; defects; flooring; finishes; composites; metals; insulation; coatings.	7 L + 4 T
III	Components; substructure/superstructure; load-bearing/non-load-bearing; structural elements; shallow and deep foundations.	8 L + 4 T
IV	Masonry; brick sizes; English/Flemish/rat-trap/header/stretcher bonds; formwork; scaffolding; lifts/stairs/escalators; roof classification and types.	7 L + 4 T

LEARNING ROADMAP

How the modules connect

1. Materials

Properties and sustainability guide selection.

2. Concrete and finishes

Ingredients and process become usable building surfaces.

3. Building system

Components transfer load and enclose space.

4. Construction technology

Masonry, temporary works, roofs and circulation make the system buildable.

MODULE 1

Fundamentals of Building Materials

Classify materials and evaluate their properties, defects, uses and sustainability.

7 L + 4 T	**CO1 / Understand**
official hours	CO/Bloom

Learning goals

- Compare natural, artificial, engineered and green materials.
- Identify stone, timber, bamboo and masonry-unit behaviour.
- Make a property-based material selection.

Key facts

- Properties are use-dependent.
- Moisture is a major durability factor.
- Sustainable choice includes life-cycle thinking.

1. Classification of building materials

Building materials may be classified by origin, manufacture and sustainability.

Concept and explanation

Natural materials occur in nature with limited processing; artificial materials are manufactured; engineered materials are designed for specific performance; green materials reduce environmental impact across their life cycle. Classification helps an engineer select a material rather than memorise isolated names.

Study points

- Natural: stone, timber and bamboo.
- Artificial: bricks, cement, glass and steel.
- Engineered/composite: plywood, MDF, particle board and reinforced concrete.
- Green selection considers local availability, durability, embodied energy and reuse.

Engineering or professional relevance

A material schedule links performance, cost, local skills, maintenance and environmental impact. The best material is context-dependent, not always the newest material.

Broad classification

Class	Examples	Selection question
Natural	Stone, timber, bamboo	How much processing and maintenance?
Artificial	Brick, cement, steel	What controlled property is required?
Engineered	Plywood, MDF, RCC	What composite performance is needed?
Green	Low-impact blocks, recycled materials	What is the life-cycle impact?

Common mistakes: Treating “natural” as automatically sustainable, ignoring durability, and comparing materials without a stated property.

Malayalam support: Building material-□□□ natural, artificial, engineered, green □□□□□□□□□□ □□□□□□□□□□□□. Selection-□ strength □□□□□□□□□ durability□□□ environment impact-□□ □□□□□□□.

2. Stone: types, properties, defects and uses

Stone is a natural material used for foundations, walls, flooring, cladding and landscape work.

Concept and explanation

Igneous, sedimentary and metamorphic stones differ in formation and therefore texture, strength, porosity and weathering behaviour. Good building stone should have adequate strength, durability, low porosity where required, workable texture and resistance to fire or chemical action appropriate to the use.

Study points

- Check grain, cracks, cavities and weathered faces.
- Match dense stone to heavy-duty use and suitable finish stone to flooring or cladding.
- Consider quarrying, transport, dressing and waste.

Engineering or professional relevance

Stone selection affects foundation performance, façade durability, floor wear and maintenance. Local stone may reduce transport but still needs testing and proper detailing.

Common mistakes: Selecting by appearance alone, ignoring water absorption, and using a porous stone in an exposed wet location without protection.

Malayalam support: Stone-□□□□ type, density, porosity, strength □□□□□□ use-□□□□ match □□□□□□□. Crack □□□□ stone structural use-□ □□□□□□□□□□.

3. Timber: properties, defects, seasoning and uses

Timber is a workable, renewable material whose performance depends strongly on species, grain, moisture and treatment.

Concept and explanation

Useful properties include strength-to-weight ratio, workability, appearance and thermal behaviour. Defects may arise from growth, conversion, seasoning or biological attack. Seasoning reduces excess moisture and improves dimensional stability; methods may be natural or artificial. Preservation protects against insects, fungi and moisture.

Study points

- Inspect knots, shakes, checks, warping and decay.
- Season before using timber where dimensional stability matters.
- Protect end grain and avoid persistent dampness.

Engineering or professional relevance

Timber is used for doors, windows, roof members, formwork and interior work. In modern practice, responsible sourcing and reuse matter as much as appearance.

Timber issue and effect

Issue	Possible effect	Control
Excess moisture	Shrinkage/warping	Seasoning and detailing
Fungal attack	Decay	Dryness and preservative
Termites	Loss of section	Treatment and inspection

Common mistakes: Confusing a natural knot with structural cracking, using wet timber in a tight frame, and ignoring termite protection.

Malayalam support: Timber-□ moisture □□□□□□□□□□□□ process □□□ seasoning. Proper seasoning □□□□□□□□□□ shrinkage, warping □□□□□□□ defects □□□□□□□□.

4. Bamboo as a sustainable material

Bamboo is a fast-growing material with useful strength-to-weight characteristics when properly selected, treated and detailed.

Concept and explanation

Its hollow culms, nodes, moisture sensitivity and biological vulnerability determine how it can be used. Treatment, protection from ground moisture, suitable joints and careful storage are essential. It can serve in temporary structures, screens, furniture, roofing support and engineered products.

Study points

- Select mature straight culms without major cracks.
- Treat against insects and fungi.
- Keep joints and end cuts protected from water entry.

Engineering or professional relevance

Bamboo can reduce embodied impact where it is locally available and responsibly harvested. Its use must be based on detailing and maintenance, not on the word “green” alone.

> **Common mistakes:** Using untreated bamboo at a wet base, making a joint that splits the culm, and assuming every species has identical properties.

> **Malayalam support:** Bamboo sustainable material □□□□, □□□□□ treatment, joint detailing, moisture protection □□□□□□ □□□□□□□□□□ durability □□□□□□.

5. Eco-friendly masonry blocks

Clay bricks, fly ash bricks, AAC blocks and compressed stabilised earth blocks (CSEB) are masonry units with different density, strength, insulation and production characteristics.

Concept and explanation

Clay bricks are familiar and durable when well fired. Fly ash bricks use an industrial by-product and have consistent shape. AAC blocks are lightweight and offer thermal benefits but need compatible mortar and careful handling. CSEB uses compacted earth with stabiliser and is sensitive to soil selection and moisture detailing.

Study points

- Compare density, compressive strength, water absorption, thermal behaviour and local availability.
- Use the correct mortar, joint thickness and curing practice.
- Do not assume a lighter block can replace a load-bearing unit without design verification.

Engineering or professional relevance

Block selection influences dead load, speed, thermal comfort, waste and transport. A comparison matrix makes the choice transparent.

Masonry-block comparison

Unit	Main advantage	Main caution
Clay brick	Established practice and durability	Quality varies with firing and absorption
Fly ash brick	Uniform shape and by-product use	Check curing and source quality
AAC block	Low dead load and insulation	Use compatible details and handling
CSEB	Local earth and low-energy potential	Protect from moisture and verify soil/stabiliser

Common mistakes: Comparing only unit price, mixing up block size with strength, and leaving porous units unprotected in driving rain.

Malayalam support: AAC block light weight □□□; fly ash, clay brick, CSEB □□□□□□□□□□ properties □□□□□□□□□□□□□. Site condition □□□□□□□□□□□□ selection.

1A. Stone testing and selection

Material properties are measured or observed because visual appearance alone is insufficient.

Concept and explanation

Selection may consider compressive strength, hardness, toughness, durability, water absorption, porosity, fire resistance and workability. The relevant property depends on whether stone is used in a foundation, wall, floor, façade or landscape.

Study points

- Write the intended use before listing properties.
- Distinguish strength from durability.
- Include water and weather exposure in the selection note.

Engineering or professional relevance

A property table supports a defensible material specification.

> **Common mistakes:** Listing every property without linking it to use, and confusing hardness with toughness.

> **Malayalam support:** Material property-□□ use □□□□□□□□□□□ select □□□□□□□□. Strength, durability, absorption □□□□□□□ □□□□□□□□□□□□□ □□□□□□□.

1B. Timber conversion and defects

Conversion changes a log into usable sections; defects affect appearance, strength and dimensional stability.

Concept and explanation

Knots, shakes, checks, warping, decay and insect attack should be identified from their cause and effect. Conversion direction and seasoning influence how defects appear or progress.

Study points

- Pair each defect with a likely cause.
- Draw a small section sketch for a defect when possible.
- Specify prevention or treatment.

Engineering or professional relevance

Defect diagnosis helps in inspection of doors, windows, roof members and formwork.

Common mistakes: Calling every line a crack and recommending paint as a cure for structural decay.

Malayalam support: Timber defect-□□□□ cause □□□□□□□□□□□□ treatment □□□□□. Crack, knot, warping □□□□□□ mix □□□□□□□□□.

1C. Clay, fly ash, AAC and CSEB production

Masonry-unit properties begin in production: raw material, forming, curing/firing and finishing. Cement-related syllabus terms include OPC, rapid-hardening cement and quick-setting cement; these are distinguished by their setting or strength-development behaviour.

Concept and explanation

Clay bricks depend on shaping and firing; fly ash bricks use an industrial by-product and controlled curing; AAC uses a lightweight cellular process; CSEB compacts selected earth with stabiliser. Cement selection must be recorded separately from masonry-unit selection. The process affects density, strength, absorption and size accuracy.

Study points

- Compare production route, density and moisture sensitivity.
- Mention curing or firing as appropriate.
- Link process to site handling and mortar.

Engineering or professional relevance

Understanding production makes test results and site behaviour easier to interpret.

Common mistakes: Treating every block as fired clay, and ignoring curing or stabilisation.

Malayalam support: Block type-□□□□ production process properties-□□ □□□□□□□□□□□□. Clay firing, fly ash curing, AAC cellular structure, CSEB stabilisation □□□□□□□□□□□□.

1D. Material sustainability and life-cycle thinking

Sustainability is a comparison of impacts over extraction, manufacture, transport, use, maintenance and end of life.

Concept and explanation

A local material may reduce transport but require more maintenance; a durable manufactured product may have higher embodied energy but longer life. Students should state the boundary and assumptions of a comparison rather than use a slogan.

Study points

- Consider durability and replacement frequency.
- Include waste, reuse and local skills.
- Separate embodied impact from operational energy.

Engineering or professional relevance

This approach supports responsible specification and avoids greenwashing.

> **Common mistakes:** Calling a material green because it is natural, and ignoring maintenance or transport.

> **Malayalam support:** Sustainable selection-□ life cycle, durability, transport, reuse, maintenance □□□□□□ □□□□□□□□□□□.

> **Module tip:** Link definitions to one worked example and one application before attempting long-answer questions.

MODULE 2

Concrete Materials and Building Finishes

Connect ingredients, concrete process, finishing materials and performance.

7 L + 4 T	CO2 / Understand
official hours	CO/Bloom

Learning goals

- Explain cement, aggregates, PCC/RCC and grades.
- Relate water–cement ratio to workability and durability.
- Differentiate flooring, boards, metals, insulation and coatings.

Key facts

- Mixing, placing, compaction and curing are a chain.
- Finishes depend on substrate preparation.
- Thermal and acoustic properties are distinct.

6. Cement types and properties

Cement is a binding material that reacts with water and helps form hardened paste, mortar and concrete.

Concept and explanation

The syllabus includes ordinary Portland cement, rapid hardening cement, quick setting cement and white cement. Selection depends on strength development, setting time, appearance and site conditions. Storage must prevent moisture entry because premature hydration reduces quality.

Study points

- Keep bags off the floor and use older stock first.
- Distinguish rapid hardening from quick setting: early strength and setting time are different ideas.
- Use white cement where colour and finish are important, subject to specification.

Engineering or professional relevance

Cement type affects construction speed, finishing, repair and appearance. It should be chosen with the mix, weather, curing and required performance.

Common mistakes: Using a special cement without a reason, storing bags in damp conditions, and confusing setting with hardening.

Malayalam support: Cement □□□□□□□□□□□ reaction □□□□□□ paste □□□□□□□□□□□□□. quick setting □□□□□□ setting time □□□□□; rapid hardening □□□□□□ early strength □□□□ □□□□□□□□.

7. Concrete ingredients, PCC, RCC and grades

Concrete combines cement, water, fine aggregate and coarse aggregate; the hardened material develops useful compressive strength.

Concept and explanation

PCC is plain cement concrete without reinforcing steel. RCC combines concrete with reinforcement so the composite member can resist tension and compression more effectively. M15, M20 and M25 are nominal grade designations in the syllabus; grade choice must follow the specification and structural requirement.

Study points

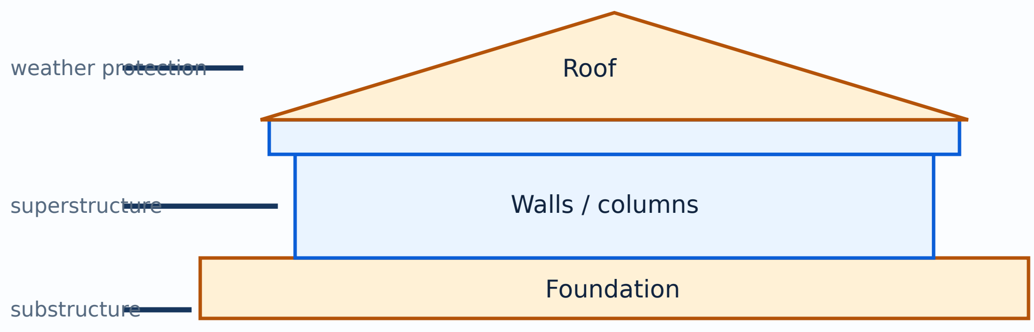
- Fine aggregate fills voids between coarse aggregate.
- Water is required for hydration and workability, but excess water can reduce strength and durability.
- RCC needs correct bar placement, cover, compaction and curing.

Engineering or professional relevance

Concrete is used in foundations, slabs, beams, columns, pavements and frames. Understanding ingredient roles prevents “add more water” site habits.

Concrete forms

Term	Meaning	Typical context
PCC	Plain cement concrete	Levelling, bedding, non-reinforced work
RCC	Reinforced cement concrete	Structural members with reinforcement
M15/M20/M25	Concrete grade designations	Select as specified for the work



A simplified vertical section from foundation to roof.

Common mistakes: Treating concrete grade as a mix recipe, adding unmeasured water, and omitting compaction or curing.

Malayalam support: PCC-□ reinforcement □□□□; RCC-□ steel reinforcement □□□□□. M15, M20, M25 grade names □□□; exact mix design specification □□□□□□□□□□□□.

8. Mix proportions and water–cement ratio

Mix proportion expresses the relative quantities of cementitious material, aggregate and water; the water–cement ratio strongly influences concrete behaviour.

Concept and explanation

More water may improve immediate workability but can increase capillary voids after evaporation. Too little water makes mixing, placing and compaction difficult. The correct balance depends on materials, placement method, exposure and required performance. Do not treat a classroom ratio as a universal design mix.

Study points

- Measure materials consistently.
- Add water gradually and observe workability.
- Protect the mix from segregation and excessive bleeding.

Engineering or professional relevance

At site, water–cement discipline is one of the simplest quality controls. It affects strength, permeability, shrinkage and durability.

Given: cement = 50 kg; water = 25 kg.
Required: water–cement ratio.
Calculation: w/c = 25/50 = 0.50.
Answer: water–cement ratio = 0.50 by mass.

Water–cement ratio calculator

Cement (kg)

50

Water (kg or L approximately)

25

Common mistakes: Adding water to make stiff concrete flow, comparing ratios without units, and ignoring aggregate moisture.

Malayalam support: Water–cement ratio = water mass / cement mass. Water □□□□□□□□□□ workability □□□□□, □□□□□ strength/durability □□□□□□.

9. Mixing, placing, compaction, curing and defects

Concrete quality depends on the complete process, not ingredients alone.

Concept and explanation

Mixing distributes cement paste and aggregates; placing puts concrete in its intended form; compaction removes trapped air; curing maintains moisture and temperature for hydration. Defects may include honeycombing, cracks, segregation, bleeding and surface dusting. Each defect suggests a different cause and remedy.

Study points

- Place near the final position to reduce segregation.
- Compact adequately without over-vibration.
- Start curing appropriately and protect fresh concrete from damage.

Engineering or professional relevance

A process checklist helps site teams find whether a defect began in batching, transport, placing, compaction, formwork or curing.

Defect and likely cause

Defect	Possible cause	Control
Honeycombing	Poor compaction or congested reinforcement	Correct placing and vibration
Segregation	Excessive handling or water	Controlled mix and placing
Cracking	Shrinkage, restraint or poor curing	Curing, joints and detailing
Bleeding	Water rising to surface	Mix control and finishing timing

Common mistakes: Assuming curing is optional, vibrating excessively, and patching honeycombing without investigating the cause.

Malayalam support: Concrete work □□□ sequence □□□: mixing → placing → compaction → finishing → curing. □□□ □□□□□□□□ quality-□□ □□□□□□□□□□□□.

10. Flooring and finishing materials

Flooring receives wear and must provide a safe, clean and visually appropriate surface. Finishes protect surfaces, correct minor irregularities and influence maintenance.

Concept and explanation

The syllabus includes stone, tile, cement and terrazzo flooring; plaster, putty and surface finishes; plywood, MDF and particle board; metals; insulation; coatings and paints. Selection considers traffic, moisture, slip, cleaning, thermal and acoustic requirements.

Study points

- Prepare the substrate before applying a finish.
- Provide joints and levels where required.
- Choose paint or coating for the exposure and substrate.

Engineering or professional relevance

A good finish is the result of material compatibility, surface preparation and workmanship. A visually attractive finish can still fail if moisture is trapped beneath it.

Common mistakes: Applying putty on a damp wall, choosing a slippery floor for a wet area, and treating MDF as weatherproof timber.

Malayalam support: Finish materials-□□□□ performance substrate preparation, moisture, traffic, cleaning □□□□□□□□□ □□□□□□□□□□□□□□□.

11. Composite boards, metals, insulation and coatings

Modern construction uses composite boards and metals to achieve speed, lightness, strength or finish quality.

Concept and explanation

Plywood uses veneers with crossed grain; MDF uses fine fibres; particle board uses wood particles and binder. Steel supplies strength and aluminium supplies lightness and corrosion resistance in suitable environments. Thermal insulation reduces heat flow; acoustic materials reduce sound transmission or reflection. Paints and coatings provide protection and appearance.

Study points

- Keep board products dry and seal exposed edges when required.
- Separate dissimilar metals where galvanic action is possible.
- Distinguish thermal insulation from acoustic treatment.

Engineering or professional relevance

Material combinations support interior partitions, doors, façades, services and energy-conscious buildings. Selection should include maintenance and replacement conditions.

Common mistakes: Using a moisture-sensitive board outdoors, assuming aluminium never corrodes, and treating a thin decorative paint as waterproofing.

Malayalam support: Thermal insulation heat flow □□□□□□□□□□□□□; acoustic treatment sound behaviour □□□□□□□□□□□□□□□□□. □□□□□□ □□□ function □□□□.

2A. Aggregate grading and workability

Aggregate size distribution affects voids, paste demand, workability and compactness.

Concept and explanation

Fine aggregate fills spaces between coarse aggregate; a balanced grading can reduce voids and improve economy. Workability is the ease of mixing, placing and compacting without segregation. The water-cement ratio is a critical control variable: adding water changes workability but can reduce strength and durability when not controlled.

Study points

- State the role of fine and coarse aggregate.
- Relate grading to paste demand.
- Use compaction and admixture ideas carefully without replacing the official syllabus focus.

Engineering or professional relevance

Aggregate selection influences pumpability, finish and strength in concrete members.

> **Common mistakes:** Calling all aggregate “sand”, and using water as the only workability control.

> **Malayalam support:** Aggregate grading voids, workability, paste demand □□□□□□□□ □□□□□□□□□□□□. Water □□□□□□ □□□□□□□□□□□ safe solution □□□□.

2B. PCC versus RCC and reinforcement role

PCC and RCC differ in structural intent and reinforcement.

Concept and explanation

Concrete is strong in compression; reinforcement supplies tensile capacity and controls cracking when correctly placed and anchored. PCC may serve as levelling, bedding or non-reinforced work, while RCC forms structural members subject to design.

Study points

- Label concrete, reinforcement, cover and formwork in a section.
- Do not assume every concrete layer is structural.
- Explain that design values come from structural specification.

Engineering or professional relevance

This distinction helps students read drawings and avoid removing or relocating reinforcement on site.

Common mistakes: Calling RCC simply “concrete with more cement”, and omitting cover or bar position.

Malayalam support: RCC-□ steel reinforcement tensile behaviour-□□ □□□□□□□□□□□□□. Bar position, cover, anchorage design □□□□□□□□□□□□.

2C. Concrete placing and construction joints

Placing should maintain the intended position and reduce segregation; joints are planned interruptions rather than accidental weak spots.

Concept and explanation

Concrete should be deposited close to its final position, placed in suitable layers and compacted. Construction joints need a planned location, clean interface and continuity of reinforcement or treatment as specified.

Study points

- Avoid long free drops.
- Compact each layer before the next.
- Record joint location and preparation.

Engineering or professional relevance

Process control protects beams, slabs, columns and foundations from honeycombing and cold joints.

> **Common mistakes:** Creating an unplanned joint at a high-stress region, and leaving laitance or debris at a joint.

> **Malayalam support:** Concrete place □□□□□□□□□□□ segregation □□□□□□□□□□. Joint planned, clean, prepared □□□□□□□□□□.

2D. Curing methods and protection

Curing maintains moisture and temperature so cement hydration can continue and the surface is protected during early age.

Concept and explanation

Water curing, covering and other specified methods are selected according to member, weather and site conditions. Curing begins after finishing at an appropriate time and continues for the required period in the project specification.

Study points

- Protect from rapid drying, rain damage and vibration.
- Keep surfaces uniformly moist where water curing is used.
- Record curing start and method.

Engineering or professional relevance

Good curing improves strength development, abrasion resistance and durability.

> **Common mistakes:** Starting too late, curing only the top surface, and treating a wet appearance as proof of adequate curing.

> **Malayalam support:** Curing hydration □□□□□□ □□□□□□□□□□□□□. Surface □□□□□□ wet □□□□□□□ complete member-□□□□ curing □□□□□□□□□□□□.

> **Module tip:** Link definitions to one worked example and one application before attempting long-answer questions.

MODULE 3

Building Components & Foundation Systems

Read a building as a load-transfer and enclosure system.

8 L + 4 T	**CO3 / Understand**
official hours	CO/Bloom

Learning goals

- Identify substructure and superstructure components.
- Differentiate load-bearing and non-load-bearing elements.
- Compare shallow and deep foundations.

Key facts

- Foundation selection depends on soil and load.
- A load path must be continuous.
- Drawings should be labelled and dimensionally sensible.

12. Building components and the building system

A building is an interconnected system of substructure, superstructure, enclosure, circulation and services.

Concept and explanation

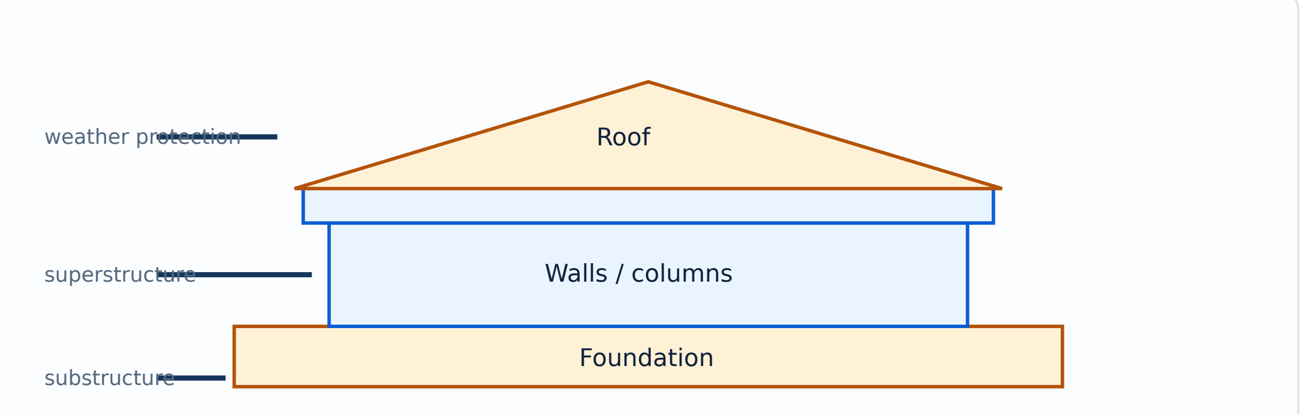
Components include foundations, walls, columns, beams, slabs, doors, windows, staircases, roofs and finishes. Load-bearing elements transfer load through a continuous path; non-load-bearing partitions mainly divide space. The architect or engineer must coordinate structure, climate response, movement, drainage and maintenance.

Study points

- Trace the load path from roof/slab to foundation and soil.
- Separate structural, enclosure and finish functions.
- Label components clearly in a building section.

Engineering or professional relevance

Understanding the system prevents isolated decisions such as selecting a wall without considering its foundation, openings, damp protection or services.



A simplified vertical section from foundation to roof.

Common mistakes: Calling every wall load-bearing, omitting damp or weather protection, and drawing components without a load path.

Malayalam support: Building system-□ foundation □□□□ roof □□□ load path □□□□□□□□□□□□□□. Wall, column, beam, slab □□□□□□□□□□ roles □□□□□□□□□□□□□.

13. Shallow and deep foundations

A foundation transfers building loads to soil safely and controls settlement.

Concept and explanation

Shallow foundations transfer load near the ground surface and include strip, isolated, combined and raft footings. Deep foundations such as piles and piers transfer load to deeper strata or develop resistance along their length and base. Selection depends on soil, load, water, settlement, neighbouring structures and construction access.

Study points

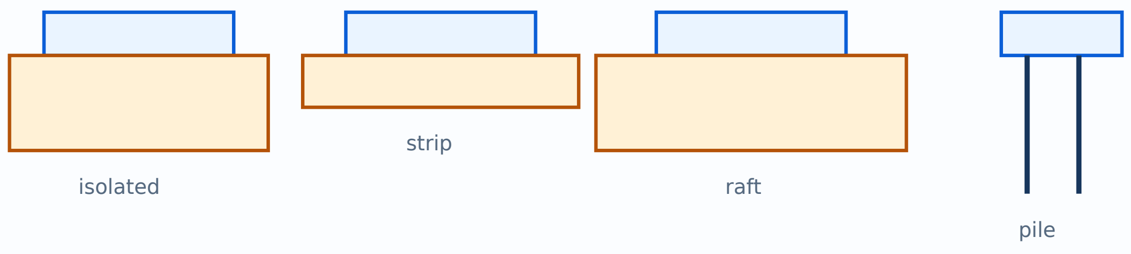
- Use isolated footings for suitable individual column loads and soil.
- Use combined or raft arrangements when spacing or soil pressure requires them.
- Use piles or piers when shallow bearing is inadequate or deep resistance is needed.

Engineering or professional relevance

Foundation choice is a site decision: a drawing alone cannot replace soil investigation and structural design.

Foundation comparison

Type	Arrangement	Typical selection idea
Strip	Continuous strip under wall	Continuous wall load
Isolated	Individual footing	Separate column load
Combined	One footing for two or more columns	Columns close or eccentric
Raft	Large common slab	Low bearing capacity or closely spaced columns
Pile/pier	Deep load-transfer element	Weak near-surface soil or deep support



Schematic comparison of isolated, strip, raft and pile foundations.

Common mistakes: Selecting a foundation by building height alone, ignoring eccentricity, and assuming a raft is always economical.

Malayalam support: Foundation load soil-□□□□□□ □□□□□□□□□□□□ transfer □□□□□□□□□□. Soil condition, settlement, water table □□□□□□ selection-□ □□□□□□□□□□□.

3A. Site investigation and foundation choice

Foundation selection begins with information about soil, groundwater, load and neighbouring conditions.

Concept and explanation

A simple decision sequence is: identify soil bearing behaviour, estimate load and settlement concerns, check site constraints, then compare shallow and deep systems. Detailed design belongs to qualified structural/geotechnical practice; the lesson teaches the selection logic.

Study points

- State the soil assumption.
- State the load and settlement concern.
- Explain why the selected foundation is suitable.

Engineering or professional relevance

This prevents a purely visual choice of isolated, raft or pile footing.

Common mistakes: Choosing by building height alone, and ignoring groundwater or adjacent foundations.

Malayalam support: Foundation select □□□□□□□ soil, groundwater, load, settlement, site constraint □□□□□□ □□□□□ □□□□□□□□.

3B. Load-bearing and framed construction

Load-bearing construction transfers loads through walls; framed construction transfers major loads through columns and beams, allowing more flexible partitions.

Concept and explanation

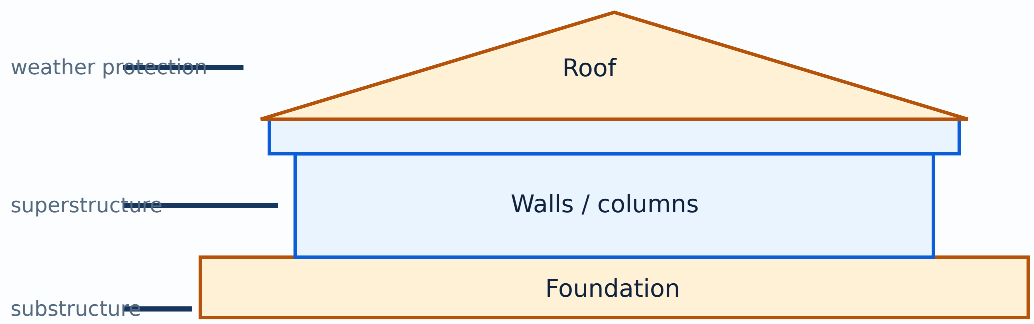
The distinction affects wall thickness, opening size, floor layout and alteration risk. A non-load-bearing partition may still carry its own weight and services, but it should not be removed without checking drawings and professional advice.

Study points

- Trace roof/slab load in each system.
- Label walls, columns and beams.
- State the effect of openings.

Engineering or professional relevance

Understanding the structural system is essential before planning doors, windows, services or interior changes.



A simplified vertical section from foundation to roof.

Common mistakes: Calling a thick wall automatically load-bearing, and treating a partition as safe to remove.

Malayalam support: Load-bearing wall load carry □□□□□□□; framed system-□ beam/column □□□□□□ load path □□□. Drawing verify □□□□□□□.

3C. Damp protection and weathering

Building components must resist rising damp, rain penetration and surface weathering.

Concept and explanation

Protection may involve plinth details, damp-proof courses, roof overhangs, flashing, sealants, slope and drainage. The correct measure depends on the source and direction of water. A finish cannot substitute for a missing water-control detail.

Study points

- Identify water source: ground, rain, plumbing or condensation.
- Show a drainage path.
- Keep porous materials protected where exposure is high.

Engineering or professional relevance

Moisture control extends material life and protects indoor comfort and finishes.

Common mistakes: Painting over damp without fixing the source, and blocking a drainage route.

Malayalam support: Moisture problem-□□□□ source □□□□□□□□□□□ finish □□□□□□ □□□□□□□□□ defect □□□□□□ □□□□.

3D. Doors, windows and openings

Openings provide access, light, ventilation, view and emergency egress while interrupting walls.

Concept and explanation

Selection considers size, orientation, security, weather, frame material, sill, lintel, threshold and operation. Structural and non-structural roles must be coordinated. A clear elevation and section show head, sill and frame relationships.

Study points

- Label head, sill, jamb, frame and shutter.
- Consider rain and ventilation.
- Coordinate lintel or beam above an opening.

Engineering or professional relevance

Openings connect building performance with human use and environmental response.

> **Common mistakes:** Drawing an opening without a lintel, ignoring swing direction and placing windows without ventilation logic.

> **Malayalam support:** Door/window opening-□ frame, sill, lintel, ventilation, rain protection □□□□□□ □□□□□□□□□□□□.

3E. Stairs, lifts and escalators

Vertical circulation is a coordinated building component, not a decorative afterthought.

Concept and explanation

Stairs require consistent rise and tread, landings, handrails, safe width and clear headroom. Lifts require a shaft, machine/service coordination and accessible entry; escalators manage continuous movement but need safety clearances and adjacent stairs or alternatives.

Study points

- Draw direction arrows and level names.
- Check rise/tread consistency in a stair sketch.
- Provide a safe alternative when escalators are not suitable.

Engineering or professional relevance

Circulation affects accessibility, fire escape, crowd movement and daily comfort.

> **Common mistakes:** Drawing stairs without direction, forgetting landings and treating a lift as a freestanding box.

> **Malayalam support:** Vertical circulation-□ safety, access, width, landing, headroom □□□□□□ □□□□□□□□□□□□.

> **Module tip:** Link definitions to one worked example and one application before attempting long-answer questions.

MODULE 4

Masonry, Roofing & Construction Technology

Apply bond, temporary-work, roofing and circulation knowledge to basic building situations.

7 L + 4 T	CO4 / Apply

official hours	CO/Bloom

Learning goals

- Explain masonry units and bonds.
- Compare formwork and scaffolding controls.
- Classify roofs and coordinate vertical circulation.

Key facts

- Staggered joints support bonding.
- Temporary works require inspection.
- Roof drainage is as important as roof shape.

14. Masonry, brick sizes and bonds

Masonry joins stone, brick or block units with mortar to form walls and other elements.

Concept and explanation

Good masonry depends on unit quality, bond, joint thickness, alignment, plumb, curing and protection. Standard brick dimensions should be treated as official dimensional information from the course context; in practice verify the specified local standard. English, Flemish, header, stretcher and rat-trap bonds arrange units for strength, appearance, cavity or material economy.

Study points

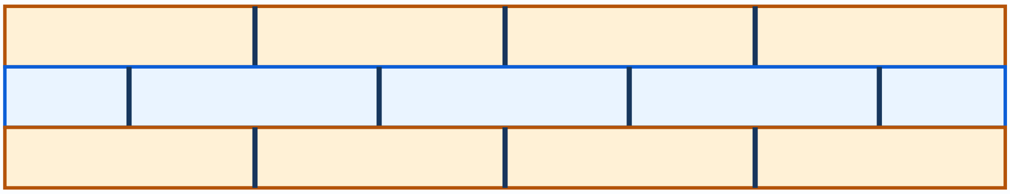
- Keep courses level and joints staggered where the bond requires it.
- Check corners and openings before filling the wall.
- Select bond according to wall thickness, structural role and finish.

Engineering or professional relevance

Bond patterns affect load transfer, continuity, material use and façade appearance. A labelled elevation is often the clearest examination answer.

Bond examples

Bond	Visual idea	Study emphasis
English	Alternate header and stretcher courses	Strength and regular arrangement
Flemish	Headers and stretchers in the same course	Appearance and bonding
Stretcher	Long face visible	Half-brick partition context
Header	Short face visible	Thicker wall/curved work contexts
Rat-trap	Bricks on edge forming cavity	Material economy and cavity



Vertical joints are staggered for bonding

A schematic elevation showing alternate courses and overlapping vertical joints.

Common mistakes: Drawing a bond without showing alternate courses, aligning all vertical joints, and confusing a brick face with a bond pattern.

Malayalam support: Masonry bond-□ units □□□□□□□ overlap □□□□□□□□□□ □□□□□□□□□ □□□□□□□□. Alternate courses-□□□□ joint alignment □□□□□□□□□□□□□.

15. Formwork and scaffolding

Formwork temporarily supports fresh concrete until it gains sufficient strength; scaffolding provides safe access and working platforms.

Concept and explanation

Formwork must be strong, stiff, stable, correctly aligned and sufficiently tight to prevent grout leakage. It should be removable without damaging the concrete. Scaffolding must provide a stable platform, safe access, bracing and edge protection. Materials, uses, merits and demerits should be compared rather than listed only.

Study points

- Check support, level, line, plumb, release treatment and openings before concreting.
- Inspect scaffold base, ties, braces, platform and guardrails.
- Do not overload temporary works.

Engineering or professional relevance

Temporary works are safety-critical even though they may disappear from the finished building. Planning them reduces accidents, rework and dimensional defects.

Temporary work

Item	Main purpose	Key control
Formwork	Shape and support fresh concrete	Strength, line, level, stripping time
Scaffolding	Access and working platform	Base, bracing, ties, guardrails

Common mistakes: Removing formwork too early, placing a scaffold on unstable ground, and treating a guardrail as optional.

Malayalam support: Formwork temporary concrete support □□□; scaffolding worker-□□ safe access/platform □□□□□□□□. □□□□□□□□□□□□ stability □□□□□□□□□□□□.

16. Roofing and vertical circulation

Roofs protect against weather and influence drainage, ventilation, heat gain and architectural character. Vertical circulation connects levels through stairs, lifts and escalators.

Concept and explanation

Roofs may be classified by shape, material and structural system. Flat, pitched, curved and trussed roofs distribute load and shed water differently. Stairs need safe rise, tread, width, handrail and landing relationships; lifts and escalators need coordinated shafts, access and safety systems.

Study points

- Match roof slope and covering to rainfall and material.
- Provide drainage and flashing at changes and penetrations.
- Coordinate stair/lift geometry with the building plan and accessibility needs.

Engineering or professional relevance

Roof and circulation decisions combine structure, climate, fire safety, movement and user comfort. A clear section often communicates more than a paragraph.

Roof types

Type	Main characteristic	Attention
Flat	Low slope surface	Drainage and waterproofing
Pitched	Inclined planes	Roof covering and rainwater flow
Curved	Continuous curved surface	Formwork and structural action
Trussed	Triangulated structural system	Member forces and connections

Common mistakes: Calling every low-slope roof “flat” without drainage, ignoring waterproofing, and drawing stairs without direction or landings.

Malayalam support: Roof shape climate, material, drainage □□□□□□□□□□□□□ □□□□□□□□□□□□□ □□□□□□□□□□□□. Stair/lift design safe movement □□□□□□□□□□□□.

4A. Brick masonry workmanship

Good brickwork begins with sound units, appropriate soaking or preparation as specified, correct mortar, line, level, plumb and curing.

Concept and explanation

The bond provides continuity and distributes load. Corners, junctions and openings need special attention. Joints should be consistent and surfaces should be protected while mortar gains strength.

Study points

- Use a gauge or level for course control.
- Check corner bonding before continuing the wall.
- Cure masonry as specified.

Engineering or professional relevance

Workmanship often determines whether a theoretically good bond performs well in practice.

> **Common mistakes:** Leaving hollow joints, building out of plumb and using damaged units at critical corners.

> **Malayalam support:** Masonry workmanship-□ line, level, plumb, joint thickness, corner bonding □□□□□□ □□□□□□□□□□□□.

4B. English, Flemish and rat-trap bond comparison

Bond names describe the arrangement of headers and stretchers across courses or within a course.

Concept and explanation

English bond alternates courses; Flemish bond alternates headers and stretchers in the same course; rat-trap bond creates a cavity by placing bricks on edge. Header and stretcher bonds emphasise the visible face and wall thickness context.

Study points

- Draw at least two courses, not one isolated brick.
- Use a legend for header and stretcher.
- State the wall thickness or cavity implication.

Engineering or professional relevance

Comparison answers score better when they show arrangement, use, appearance and a limitation.

Bond answer checklist

Item	What to show
Arrangement	Course or within-course pattern
Function	Bonding, appearance or cavity
Detail	Header/stretcher legend
Caution	Wall thickness and workmanship

Common mistakes: Drawing a pattern that does not repeat, and using “English” or “Flemish” as unexplained labels.

Malayalam support: Bond answer-□ alternate course, header/stretcher arrangement, use, limitation □□□□□□ □□□□□□□□.

4C. Formwork requirements and stripping

Formwork carries fresh concrete, construction loads and pressure until the concrete is sufficiently stable.

Concept and explanation

It should be designed or selected for strength, stiffness, stability, line, level, plumb, tightness, access and safe removal. Stripping time depends on member, span, support and concrete development; follow the project specification rather than a guessed number.

Study points

- Check props and bracing.
- Apply release treatment without contaminating reinforcement.
- Inspect before pour and after removal.

Engineering or professional relevance

Formwork quality controls member dimensions, surface finish and worker safety.

> **Common mistakes:** Stripping too early, moving props without approval and leaving grout-leak paths.

> **Malayalam support:** Formwork remove [illegible] [illegible] concrete strength, span, support condition [illegible] [illegible].

4D. Scaffolding safety and types

Scaffolding creates a temporary working platform at height and must be stable, accessible and protected.

Concept and explanation

Types may be compared by material, support and arrangement. The common principles are sound foundation, standards, ledgers, bracing, ties, platform, guardrails, toe boards and safe access. Inspection is required after alteration or adverse weather.

Study points

- Keep platforms fully boarded and uncluttered.
- Provide guardrails and toe boards.
- Prevent unauthorised alteration or overloading.

Engineering or professional relevance

Safe temporary access directly affects masonry, plastering, painting and maintenance quality.

> **Common mistakes:** Using loose supports, missing braces and allowing material storage to overload a platform.

> **Malayalam support:** Scaffolding-□ base, bracing, tie, platform, guardrail, toe board, access □□□□□□ □□□□□□□□□□□□.

4E. Roof drainage and waterproofing

A roof must shed water and protect the interior; shape alone does not guarantee performance.

Concept and explanation

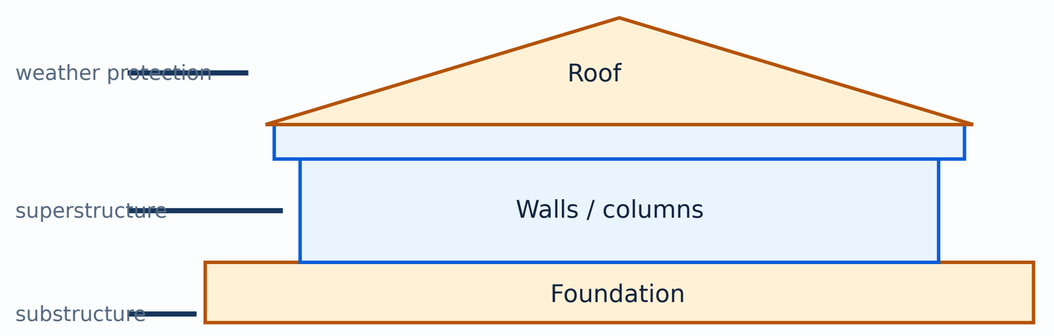
Flat roofs require slope, outlets, upstands and waterproofing. Pitched roofs require covering, overlaps, valleys, ridges and gutters. Curved and trussed roofs require coordinated structure and enclosure. Penetrations and junctions are common failure points.

Study points

- Draw the water path with arrows.
- Locate outlets at low points.
- Detail flashing at changes and penetrations.

Engineering or professional relevance

Roof detailing combines material, climate, structure and maintenance.



A simplified vertical section from foundation to roof.

Common mistakes: Calling a flat roof level, blocking outlets and ignoring waterproofing at parapets.

Malayalam support: Roof drainage-□ slope, outlet, flashing, waterproofing □□□□□□ □□□□□□□□□□ □□□□□□□□□.

4F. Selection of construction technology

Construction technology is selected by scale, material, labour, speed, safety, cost, climate and quality control.

Concept and explanation

A basic comparison can consider conventional masonry, framed construction, prefabricated or composite elements, and site constraints. The correct answer is conditional: explain why a method fits the stated situation.

Study points

- State the project constraint first.
- Compare speed, skill, material and maintenance.
- Include temporary works and safety.

Engineering or professional relevance

Technology choice is an engineering decision, not a fashion label.

Common mistakes: Ranking a method without criteria, and ignoring site logistics or worker safety.

Malayalam support: Construction technology select □□□□□□□□□□□ speed, skill, material, safety, cost, climate, quality □□□□□□ compare □□□□□□□.

Module tip: Link definitions to one worked example and one application before attempting long-answer questions.

RAPID REFERENCE

Formula and concept bank

Water–cement ratio

```
w/c = mass of water / mass of cement
```

Keep units consistent; the ratio is dimensionless.

Concrete selection

```
grade → exposure → ingredients → process → curing
```

The syllabus names M15, M20 and M25; do not treat the grade name alone as a complete mix design.

Load path

```
roof/slab → beam/column/wall → foundation → soil
```

Use arrows in a building section.

Material selection

```
performance + durability + cost + sustainability
```

State the intended use before comparing materials.

RAPID REVISION

Diagram library

Building component relationship

Trace substructure, superstructure and roof.

Foundation types

Compare isolated, strip, raft and pile foundations.

Brick bond principle

Show staggered joints and course arrangement.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Complete these activities as evidence of self-learning. Keep sketches, tables, screenshots, observations or short reports in a subject file.

1. Material classification chart

Prepare a chart classifying natural, artificial, engineered and green materials.

4 hours

2. Stone/timber/bamboo case study

Document properties, defects, treatment and a local application.

4 hours

3. Eco-block comparison

Compare clay, fly ash, AAC and CSEB blocks using a property matrix.

4 hours

4. Cement and concrete flowchart

Show cement types, ingredients, mix/w-c ratio, concreting and curing.

4 hours

5. M15/M20/M25 and PCC/RCC note

Prepare a concise grade and application comparison.

4 hours

6. Concrete process and defect record

Document mixing, placing, compaction, curing and a defect investigation.

4 hours

7. Finish-material survey

Identify flooring, plaster, putty, boards, metals, insulation and coatings.

4 hours

8. Building-component drawings

Draw and label a building section and load-bearing/non-load-bearing elements.

4 hours

9. Foundation case study

Compare two foundation systems for a stated soil and load situation.

4 hours

10. Masonry and bond illustrations

Document brick properties, sizes and English/Flemish/rat-trap/header/stretcher bonds.

4 hours

11. Formwork/scaffolding illustration

Prepare a safe temporary-works sketch with merits and demerits.

4 hours

12. Roof and circulation chart

Classify roof types and show stairs, lifts and escalators in a circulation diagram.

4 hours

13. Site observation

Record one material-selection or construction-quality observation with photographs or sketches.

4 hours

14. Sustainability reflection

Explain how durability, local sourcing and maintenance affect material choice.

4 hours

15. Revision portfolio

Assemble all tables, labelled sketches and comparison notes for assessment.

4 hours

EXAMINATION PREPARATION

Assessment methodology

The official assessment is CIE 40 and ESE 60. The displayed structure includes attendance 5, four self-learning activity components and two series examinations converted as specified in the syllabus, followed by a 60-mark ESE.

Series examination

Duration: 2 hours. The paper uses the official 1/3/7-mark structure, with module-set choices as specified in the PDF and a total of 50 marks before conversion.

ESE

Duration: 2.5 hours. Questions are distributed across all four modules using Part A 1-mark, Part B 3-mark and Part C 7-mark components for a total of 60 marks.

Exam discipline: Do not label this lesson’s model paper as an official examination paper. It is a practice aid based on the official pattern.

ACTIVE RECALL

Module-wise questions and answers

Expand each question only after attempting it from memory. Match answer length to the mark value.

Module 1

What is a green building material?

A material selected with attention to environmental impact across its life cycle, including processing, transport, use, maintenance and end of life.

Name three stone properties.

Strength, durability, water absorption/porosity and workability are relevant properties; the correct set depends on use.

Why is timber seasoned?

Seasoning reduces excess moisture and improves dimensional stability and service performance.

Why is bamboo treatment needed?

Treatment protects the material from insects, fungi and moisture-related deterioration.

Module 2

What is PCC?

Plain cement concrete without reinforcing steel.

Differentiate rapid hardening and quick setting cement.

Rapid hardening develops early strength quickly; quick setting has a shorter setting time. They are not the same property.

What is water–cement ratio?

The mass of water divided by the mass of cement in a concrete mix.

Name two concrete defects.

Honeycombing, segregation, bleeding, cracking and surface dusting are examples.

Module 3

What is a load path?

The continuous route through which loads travel from roof/slab and other components to the foundation and soil.

What is a strip footing?

A continuous shallow foundation generally used under a wall or line load.

Differentiate substructure and superstructure.

Substructure is below or near ground level, mainly foundations; superstructure is the part above the ground line.

Name two deep foundations.

Pile and pier foundations.

Module 4

Why are brick joints staggered?

Staggering improves bonding and reduces continuous weak planes.

What is formwork?

Temporary mould and support used to shape and hold fresh concrete.

What is scaffolding?

A temporary access and working platform system used during construction.

Name two roof classifications.

Roofs may be classified by shape, material or structural system; flat, pitched, curved and trussed are shape/system examples.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Compare materials by performance and context; do not use a nominal grade, bond name or roof type without explaining its purpose.

Part A — 1 mark each

Answer all short definitions and identifications.

Part B — 3 marks each

Answer concise explanations, comparisons or procedures.

Part C — 7 marks each

Answer structured long questions with diagrams, tables, examples or applications.

Self-check

Reserve the last 10 minutes to check labels, units, terminology and question choices.

FINAL REVISION

Quick revision and exam checklist

- Module 1: revise the official headings, terms and one labelled diagram.
- Module 2: revise the official headings, terms and one labelled diagram.
- Module 3: revise the official headings, terms and one labelled diagram.
- Module 4: revise the official headings, terms and one labelled diagram.
- Practise at least one 1-mark, 3-mark and 7-mark answer.
- Read every official self-learning activity as a possible viva or assignment prompt.

Common mistakes

Do not confuse definitions with examples, omit labels from diagrams, copy a formula without symbol meanings, or write unsupported claims as official syllabus information.

Last-day checklist

Revise every module heading, formula/concept bank, diagram library, activity list, assessment pattern and at least one answer per mark category.

VOCABULARY

Glossary

Important terms

Term	Short meaning

OFFICIAL RESOURCES

References

Textbook(s)

- Construction Materials — D.N. Ghose, Tata McGraw Hill; Building Construction — S.C. Rangwala, Charotar Publishing; Building Construction — Sushil Kumar, Standard Publishers.

Reference books

- Engineering Materials — S.C. Rangwala, Charotar Publishing, Ahmedabad
- Laboratory Manual on Testing of Engineering Materials — H. Sood, New Age Publishers, New Delhi
- Civil Engineering Materials — N. Subramanian, Oxford University Press

Web resources listed in the syllabus

- https://bis.gov.in (All)

IN-PAGE SEARCH

Search this lesson